

In Class Discussion

EN 601.124: The Ethics of Artificial Intelligence and Automation

Department of Computer Science, February 18th 2026





Mullainathan suggests that the Obermeyer et al. healthcare algorithm study shows how algorithms can reveal hidden biases in existing systems.

But the bias was only discovered because researchers had access to the algorithm. How realistic is it to expect this kind of auditing at scale, especially with proprietary systems?

**Biased Algorithms are
Easier to Fix than Biased
People**





**How did the developers of the algorithm
measure sickness and why did that
produce a biased output?**



Why do you believe that implicit bias training has a modest impact?



**How does the decision-making process
vary between humans and algorithms?**



Do you believe algorithms or humans are easier to fix?



Are there any other cases of algorithmic bias that could be easily fixed?

Inherent Trade-Offs in the Fair Determination of Risk Scores





What are the fairness properties for risk assessments? Do you agree or disagree with them? Why?



**What is the difference between the
calibration and balance conditions?
Which is more important?**



In your own words, describe when a risk assignment can achieve all three fairness guarantees simultaneously.



Give one real-world application of a situation that would produce all fairness guarantees. How would this be accomplished?

Synthesis





Mullainathan's argument assumes that once we detect algorithmic bias, we can fix it. But given Kleinberg et al.'s impossibility result, "fixing" bias requires choosing which fairness criterion to satisfy and which to violate.

In your opinion, who should make that choice: engineers, ethicists, legislators, or affected communities?



Both readings implicitly compare algorithms to the status quo of human decision-making.

But deploying an algorithm changes the system itself: it can create feedback loops (e.g., more policing in areas flagged as high-risk leads to more arrests, which reinforces the score).

How could we address these dynamic effects?



Both readings focus on supervised learning settings where we predict a binary outcome.

But many real-world harms come from how predictions are used downstream (e.g., a risk score informing but not determining a judge's decision).

Should fairness constraints be applied to the score itself, to the final decision, or to the outcome